vision2learn

Coding Practices Level 3

Hints and Tips

Unit 4

Contents

Hints and Tips Unit 4

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 4: Software testing

Section 1 – Understand different levels, types and methods of testing

Q1. Identify different levels, types and methods of testing.

As an identify question, the answer doesn't need to be long, but should give a short statement to identify the generally known levels, types and methods of testing.

Q2. Explain the relationship between four levels, types and methods of testing used within the stages of the Software Development Lifecycle. Within your answer, include:

- Black box and white box methods
- Where and why they are used
- Who they are used by

Your answer must include a minimum of four testing methods and explain the relationship between them. Be sure to include the bullet points within your answer for each method that you include.

Q3. Using the table below, explain four differences between functional and non-functional testing.

Using the table you should explain four separate differences between function and non-functional testing. Think about what each one does differently and how each/when each could be applied. Make sure that each difference is fully explained and that you do simply identify within your answer.

Section 2 – Know about test driven development

Q1. Explain what is meant by Test Driven Development (TDD).

You should demonstrate your understanding of what TDD (test driven approach) is and include the steps required when taking a TDD (test driven) approach.

Q2. Describe the structural composition of a unit test. Within your answer, you must include what is meant by:

- Setup
- Arrange
- Act
- Assert
- Teardown

It's important to include each of the bullet points within your description of the structural composition of a unit test.

Q3. Explain the advantages and disadvantages of unit testing.

This explanation should include the advantages and disadvantages that a TDD approach can bring. Think about when it is best applied and when it's not. Why is that?

Q4. Consider what would make a bad unit test.

Think about what elements could make a bad unit test and include your understanding of these. Consider any complications, multiple factors and specific details amongst other areas within your answer.

Section 3 - Be able to identify and fix a bug

Q1. Explain techniques which can be used to identify and fix bugs.

Within your explanation consider the different steps that are taken to identify and fix bugs. For each step that you include ensure to fully explain what's involved. You could also include the benefits of each step.

Q2. In the table below, describe the benefits of each technique, (Variable watching, compiler warnings, and Unit test), and include an explanation of the following:

- What each technique is and how it's performed
- When each technique might be helpful
- When each technique might not be helpful

For each technique in the table, your description must include an explanation of all three bullet points within the question. Ensure that your answer focuses on the benefits of each technique.

Q3. You've been given a task for this question on page 1, session 3 of the course content (see Time to practice). To complete the task you will need to diagnose the cause of the bug within the code that you've been given. Once diagnosed you will fix the broken code and upload your file with the fix.

When diagnosing the problem, you must use the following techniques:

- Variable watching
- Compiler warnings
- Unit test

For this question, you will need to follow the instructions within the Time for Assessment on page 1, session 3 of the course content. First, you should diagnose the problem within the code, ensuring that you use the three techniques bullet pointed. Once diagnosed, you will need to fix the code and create a file for your fix. Upload this file with your assessment document for your tutor to assess.

Q4. Now that you've diagnosed and fixed the bug, reflect on how the bug could have been prevented. Within your reflection include:

- What the problem was
- How you diagnosed the problem using the three techniques in question 3
- How you fixed the problem
- An explanation for how you think the problem occurred
- How the problem could have been prevented
- The steps that can be made to prevent a similar future occurrence

There's several steps to reviewing the coding bug and you should check carefully that you've included all of the detail requested within the bullet points. In particular, it's important that you include how you diagnosed the problem and reference the three techniques from question 3.